The Asymmetric Effect of Information Shock on Overnight and Intraday Expected Returns: Evidence from Chinese A-share Stock Market

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We uncover interesting opposite effects of jumps in stock prices on three kind of stock returns: close-to-close expected return, and its two components, namely overnight and intraday return, when heterogeneous investors are confronted with different types of news during trading hours and non-trading hours. In particular, on the one hand, we provide evidence that the long-short strategies based on jump return can generate statistically positive risk-adjusted overnight and close-to-close alphas, while a negative intraday alpha. On the other hand, the Fama-MacBeth regressions demonstrate a positive relation between jump return and the subsequent stock overnight/close-to-close returns, confirming that the positive risk-return tradeoff originates from the overnight period. The positive overnight and close to close premium is pervasive of the least liquid stocks and small firms and particularly microcaps. Further analysis shows that it is positive jump return rather than negative jump return that mainly contributes to the positive (negative) overnight (intraday) return due to investors' preference to buying winners in Chinese stock market.

Keywords: Jump measures; Overnight return; Intraday return; Close-to-close return; Risk-return tradeoff
JEL Classification Codes：G11；G12；G14

1. Introduction

Investors receive a wealth of information every day (Beber and Brandt, 2010). Heterogeneous investors obtain different types of news involving public news to private news, during day and at night (Boudoukh et al., 2019). The latent news process induces two separate components, normal and unusual news events, governing the distribution of individual stock returns. The unusual news causes infrequent large moves in returns, labelled as jumps (Maheu and Mccurdy, 2004). In empirical studies, a number of studies have suggested that jump in stock prices can forecast stock returns in a positive (or inverse) direction (Bollerslev et al.,2016; Qi, 2019; Bollerslev et al.,2020).

The literature widely documents investors' overreaction or underreaction to information shock, which means that the proxy of information shock is positive or negative related to the future stock returns. For example, De Bondt and Thaler (1987) and Cooper (1999) attribute a long-term reversal pattern in stock returns to investor's overreaction to information shock. Later, the pattern of short-term stock return momentum is demonstrated by Jegadeesh and Titman (1993) and it has been shown that liquidity or market microstructure effects mainly contribute to investor's under-reaction to information shock (Jegadeesh and Timan, 1995; Avramov, Chordia, and Goyal, 2006; Gutierrez and Kelley, 2008). In these studies, information shocks are usually driven by some observable specific corporate events, such as the post-earning announcement drift (Ball and Brown, 1968; Foster, Olsen and Shevlin, 1984), dividend initiations and omissions (Michaely, Thaler, and Womack, 1995) and stock split announcements (Ikenberry and Ramnath, 2002). However, Jiang and Zhu (2017) first use the jump in stock prices as a

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proxy for information shock and attribute the short-term stock return momentum in the US equity market to investor's under-reaction.

In recent years, the literature documents price movements during trading hours and non-trading hours (Aboody et al., 2018; Laurence et al., 2018; Lou et al., 2019; Hendershott et al., 2020; Bogousslavsky, 2021). More Specifically, Aboody et al. (2018) propose that overnight return can be used as a proxy variable of individual stock sentiment. Lou et al. (2019) link the heterogeneity of investors with the persistence of overnight and intraday returns, finding profits based on reversal and a variety of momentum strategies are earned entirely overnight. Hendershott et al. (2020) show that beta is positively (negatively) associated with expected overnight (intraday) returns. Bogousslavsky (2021) provides evidence that abnormal returns gradually correct during the day but worsen at the end of the day because the margin requirements are higher overnight and the lending fee exists only at night. Moreover, the literature provides evidence that overnight returns are primarily information-driven (Jones et al., 1994; Santosh, 2016; Boudoukh et al., 2019; Barardehi et al., 2021). For instance, Boudoukh et al. (2019) conclude that fundamental information in news accounts for nearly half of overnight idiosyncratic volatility. Barardehi et al. (2021) find that overnight price movements are likely irrelevant for liquidity measurement, but largely reflect information arrival, and their modified Amihud illiquidity measure using open-to-close returns perform significantly better than the classic close-to-close return based on Amihud illiquidity.

Motivated by the literature on the investor mis-reaction to information shock and novel overnight/intraday return-risk tradeoff, in this paper, we investigate the impact of information shock on three kind of stock returns. In particular, we examine empirically the response of Chinese cross-sectional stock overnight, intraday, close-to-close returns to the jump return. We will uncover interesting opposite effects when heterogeneous investors have access to different types of news at different times (during the day and at night). Specifically, we first calculate jump measures using high-frequency data according to Barndorff Nielsen and Shephard (2004, 2006) and Huang and Tauchen (2005), assuming that the jump is rare and there exists big change in stock prices. Then, we calculate the intraday and overnight components of stock-level close to close returns according to Lou et al. (2019). Finally, taking the jump in high frequency stock prices as large information shock measurement, we examine the relation between large information shock and subsequent components of returns (that is, overnight return, intraday return, and close to close return) in the Chinese stock market by long-short portfolio analysis and Fama-MacBeth regression both at the individual stock level and portfolio level.

Our empirical results show that, on the one hand, the long-short strategies based on jump return can generate statistically positive risk-adjusted overnight and close-to-close alphas. On the other hand, the Fama-MacBeth regressions demonstrate a positive relation between jump return and the subsequent stock overnight/close-to-close returns, while a negative relation between the jump return and the subsequent stock intraday return. These findings provide the evidence that the positive traditional (close-to-close) return-jump risk tradeoff mainly comes from the non-trading period, that is, it is the overnight component rather than the intraday component of close-to-close return that contributes to the return-higher-order moment risk tradeoff. Overall, the positive return-jump tradeoff at night is robust after controlling for other firm characteristics and market investors' sentiment, and the positive overnight premium is

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pervasive of the least liquid stocks and small firms and particularly microcaps. Finally, further analysis confirms that investors show an asymmetric mis-reaction to the positive and negative information shock due to their potential preference to buying the winner in Chinese stock market. In essence, it is the positive jump return rather than negative jump return that contributes to the positive (negative) abnormal overnight/close-to-close (intraday) return.

Our paper strengthens the literature about whether jump in stock prices can explain stock returns. For instance, Yan (2011) uses the slope of implied volatility smile as a proxy for jump size and demonstrates a negative predictive relation between stock return and jump size. Zhou and Zhu (2011) examine the relation between cross-section expected excess returns and variations in jump intensities and find that jump intensity can explain the individual stock excess returns in Chinese stock market. Bollerslev et al. (2016) utilize high-frequency data set to investigate two rough betas associated with intraday discontinuous and overnight returns, that is, jump betas entail significant risk premiums. Further, Bollerslev et al. (2020) consider the differential pricing of the firm specific "good" versus "bad" volatilities might stem from overreaction to negative "jumps" and verified that the predictability of relative signed jump is stronger among small firms, firms with more volatile stock prices and more illiquid firms. Qi (2019) finds jump beta anomaly in Chinese stock market, which indicates the jump risk could negatively forecast the stock return. Our paper links the jumps measures with the components of returns from the perspective of information shocks. Different from Jiang and Zhu (2017) who use daily return data to extract jump returns, we use high-frequency 5-minute data to extract jump returns as measures of information shock, which can not only increase the length of the sample and improve the accuracy, but also enrich the interpretation of a price jump as an information event.

Our research is also related to the literature on the investor's mis-reaction to information shock, with an interesting pattern of short-term underreaction at night (that is, a positive relation between information shock measures and overnight return) but short-term overreaction during the day. In addition, we contribute to the literature on the information-driven or liquidity-driven abnormal return during the day and at night. Our results parallel the works of Hendershott et al. (2020) and Bogousslavsky (2021) that show that beta is positively (negatively) associated with expected overnight (intraday) returns.

The remainder of the paper is as follows: Section 2 describes the data, the identification of jump measures, and the calculation of components of returns. Section 3 presents the empirical analysis. Section 4 discusses the potential mechanism. Section 5 contains a further analysis. Section 6 concludes.

2. Data and methodology

2.1. data description

This paper takes the high frequency stock prices and daily frequency stock prices of Chinese A-shares as samples. The high-frequency data, Fama-French-Carhart factors and risk-free rate come from China Stock Market & Accounting Research Database (CSMAR), provided by GuoTaiAn. The daily frequency stock data is used to directly calculate the daily components of returns. Specifically, daily overnight return is defined as the logarithm of the opening price on day t minus the logarithm of the closing price on day $t-1$, and daily intraday return is defined as the logarithm closing price minus the logarithmic opening price of the same day. The sample period is from January 2010 to July 2017.

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2.2. Detection and identification of jump return

Jump is a rare and big change in stock prices, which is usually caused by unexpected information. A lot of papers use big price changes as proxy variables of information events (Conrad, Cornell, Landsman, and Rountree, 2006; Savor, 2012). In this paper, we use the methods of Barndorff-Nielsen and Shephard (2004) and Huang and Tauchen (2005) to identify jump risk variables and calculate the corresponding (positive and negative) jump returns. Then we use the portfolio analysis to empirically explore whether those information shock measures possess predictive effect on short-term overnight return and short-term intraday return.

More specifically, based on the economic intuition of the characteristics and sources of jumps in financial markets (Merton, 1973), Barndorff-Nielsen and Shephard (2004) and Huang and Tauchen (2005) assumed that jump occurs at most once a day, and when jump does occur, the size of the jump dominates the yield of the day. Under this assumption, at the significance level of $1-\alpha$, we can get the estimator of discrete realized jump volatility and its asymmetric components, that is, positive realized volatility, and negative realized jump volatility, as follows:

$$RJV_t = I\left(Z_t \geq \Phi_\alpha^{-1}\right)\sqrt{\left[RV_t - BV_t\right]} \tag{1}$$

$$Positive_RJV_t = I(r_t \geq 0)RJV_t \tag{2}$$

$$Negative_RJV_t = I(r_t \leq 0)RJV_t \tag{3}$$

Where $I_{t,\alpha} = I\left(Z_t \geq \Phi_\alpha^{-1}\right)$ and $I(r_t \geq 0)$ represent the indicator variables; Φ is the cumulative standard normal distribution function; $I(r_t \geq 0)$ equals to 1 if the daily return $r_t \geq 0$; RV_t is the realized variance; RJV_t is the daily frequency realized jump volatility. Through the above test of Z_t statistics and related quadratic variation theory, we can get the estimators of continuous sample path variance (BV_t)and discrete jump variance (RJV_t)of return volatility of financial assets. The accuracy of realized jumps depends on the assumption that there is only one jump in a trading day. Although there may be many large jumps in asset prices in a day due to market sentiment rather than information disclosure, Lee and Mykland (2008) also support the hypothesis that jumps are rare and large. The expression of Z_t statistic is as follows:

$$Z_t = \frac{\left(RV_t - BV\right)/RV_t}{\sqrt{\left(\frac{\pi^2}{4} + \pi - 5\right)\frac{1}{M}\max\left(1, \frac{TP_t}{BV_t^2}\right)}} \tag{4}$$

$$TP_t = \frac{M}{M-2}\cdot\frac{M}{4\left[\Gamma(7/6)/\Gamma(1/2)\right]^3}\cdot\sum_{i=3}^{M}\left|r_{t,i}\right|^{4/3}\left|r_{t,i-1}\right|^{4/3}\left|r_{t,i-2}\right|^{4/3} \tag{5}$$

$$RV_t = \sum_{j=1}^{M} r_{t,j}^2 \rightarrow \int_{t-1}^{t}\sigma_s^2 ds + \int_{t-1}^{t} J_s^2 q_s \tag{6}$$

Equation (6) is based on the realized jump variance calculated by Andersen and Bollerslev (1998)

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based on high-frequency 5-minute data. Then, Barndorff-Nielsen and Shephard (2004, 2006) propose the realized bi-power variation(*RBV*), that is, the continuous sample path variance(BV_t)， mentioned above and the discrete realized jump volatility(RJV_t)separated from equation (6).

Based on the realized jump volatility (*RJR*), the daily return corresponding to a single daily frequency jump is recorded and marked as jump return (*JR*), and the daily frequency return of non-jump occurrence is marked as 0. In addition, the positive and negative jump returns (*PJR* and *NJR*) are marked according to the positive and negative jump risk measures of daily frequency realized jump volatility.

2.3. Components of the close-to-close return

We calculate the intraday and overnight components of firm-level returns according to Lou et al. (2019). Specifically, for each stock *i*, the definition of intraday return is the price appreciation between market open and close of the same day *s*, while the overnight return is the price appreciation between market open price of day s and close of the day *s-1*,

$$r^i_{\text{int}raday,s} = \frac{P^i_{close,s}}{P^i_{open,s}} - 1 = \ln P^i_{close,s} - \ln P^i_{open,s} \tag{7}$$

$$r^i_{overnight,s} = \frac{P^i_{open,s}}{P^i_{close,s-1}} - 1 = \ln P^i_{open,s} - \ln P^i_{close,s-1} \tag{8}$$

$$r^i_{close-to-close,s} = \frac{P^i_{close,s}}{P^i_{close,s-1}} - 1 = \ln P^i_{close,s} - \ln P^i_{close,s-1} \tag{9}$$

In empirical analysis, we will use the daily continuous compounded log overnight return, log intraday day and log close-to-close return on the stock.

In further research, we calculate intraday return, overnight return and close-to close return of each month *t* by accumulating corresponding daily intraday return and overnight return.

$$r^i_{\text{int}raday,t} = \sum_{s \in t} r^i_{\text{int}raday,s} \tag{10}$$

$$r^i_{overnight,t} = \sum_{s \in t} r^i_{overnight,s} \tag{11}$$

$$r^i_{close-to-close,t} = \sum_{s \in t} r^i_{close-to-close,s} \tag{12}$$

3. Empirical analysis

In this section, with identified jumps in high frequency stock price, we examine the relation between large information shock measure (i.e., jump return) and subsequent components of returns in the Chinese stock market. We first implement the trading strategies by the following Fama-French-Carhart four factors model,

$$OvernightR_{P,t} - Rf_{P,t} = alpha + \beta FF3_{P,t} + \gamma Momemtum_{P,t} + \varepsilon_t \tag{13}$$

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Where, the subscript P（P = 1,2,…,10,10-1）represents the stock portfolio deciles based on the jump returns, and $OvernightR_{P,t}$ is the overnight return of the portfolio P; $Rf_{P,t}$is the risk-free rate; $FF_{P,t}$is the Fama-French three-factor; $Momemtum_{P,t}$ is the Carhart momentum factor.

We next examine Fama-MacBeth regression of cross- section of overnight, intraday, and close-to-close expected returns on lagged jump returns and other firm characteristics, respectively. Finally, we turn to describe the asymmetric impact of asymmetric information shock on overnight and intraday expected return.

3.1. Descriptive statistical analysis

Table 1 shows the basic descriptive statistics of individual stocks' overnight return/intraday/close to close return and jump returns.

Table 1

Descriptive statistics

Panel A: Descriptive statistics

	mean	std	min	25%	50%	75%	max	count
overnight return	-0.0016	0.0136	-0.1053	-0.0059	0.0000	0.0027	0.0954	2123711
intraday return	0.0017	0.0299	-0.2009	-0.0129	0.0019	0.0165	0.2006	2123711
close-to-close return	0.0001	0.0317	-0.1071	-0.0149	0.0011	0.0158	0.0974	2123711
jr	-0.0007	0.0437	-1.3982	-0.01201	0.0006	0.0118	0.1288	186091
pos_jr	0.0211	0.0240	0.0001	0.0053	0.0118	0.0262	0.1288	93438
neg_jr	-0.0225	0.0480	-1.3982	-0.0250	-0.0122	-0.0056	-0.0001	92653

Panel B: Mean jump returns of the value-weighted overnight return stock portfolio

Deciles	1	2	3	4	5	6	7	8	9	10
jr	-0.0494	-0.0185	-0.0124	-0.0078	-0.0038	0.0003	0.0048	0.0107	0.0200	0.0489
neg_jr	-0.0802	-0.0334	-0.0260	-0.0213	-0.0183	-0.0152	-0.0124	-0.0099	-0.0072	-0.0042
pos_jr	0.0039	0.0066	0.0091	0.0120	0.0155	0.0183	0.0233	0.0298	0.0403	0.0659

Note: jr, pos_jr and neg_jr are daily jump return, positive jump return and negative jump return respectively.

Panel A of Table 1 shows that the mean daily overnight return of Chinese stock market is -0.16%, while the mean daily intraday return is 0.17% and mean daily close-to-close return is 0.01%, which is consistent with previous work achieved by Qiao and Dam (2020) that has argued that the overnight return is negative due to the "T+1" trading system in Chinese stock market, see Fig1; The absolute value of the mean value and highest value of daily positive jump return (2.11% and 12.88%) are both smaller than the mean value and the smallest group (2.55% and 13.98%) of negative jump return respectively. However, Panel B of Table 1 presents that from portfolio 6 to portfolio 10, the mean lagged positive jump returns are greater than the absolute value of mean lagged negative jumping returns. These preliminarily show that, on the one hand, the overall return of Chinese stock market is not good and there are more extreme risks; on the other hand, for investors, the asymmetry of positive and negative jump returns is likely to contain favorable arbitrage opportunities.

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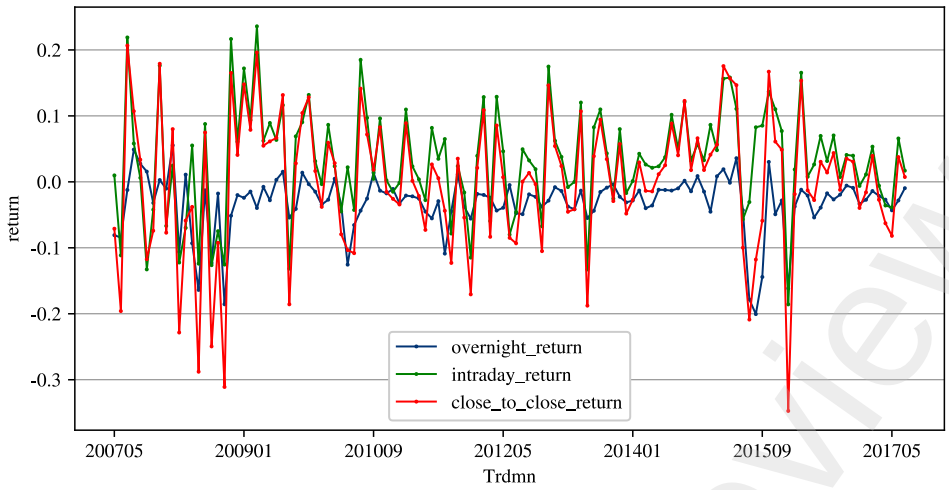


Fig1.Time series chart of average overnight return, intraday return and close-to-close return of individual stock samples during the sample period.

3.2. Information shock and components of return

We decompose the abnormal return into their intraday and overnight components. Then, we calculate the average adjusted Fama-French-Carhart alphas of the overnight and intraday components of the long-short trading strategies based on jump return.

Table 2 represents that when we employ a trading strategy of longing the high jump return decile and shorting the low jump return decile, a Fama-French-Carhart adjusted overnight premium of 1.33% with an associated *t*-statistic of 25.34 can be obtained, see Panel A, while the close-to-close Fama-French-Carhart adjusted alpha is actually slight positive (1.29% per day, *t*-statistic of 20.73), see Panel C. However, a Fama-French-Carhart adjusted intraday alpha can't be obtained through the above strategy (-0.04% per day, *t*-statistic of -1.95). That is, the jump return can positively forecast the future stock return, and this positive risk-return tradeoff originates from the overnight period. Figure 2 plots the corresponding graph.

Table 2
Abnormal overnight/intraday/close-to-close return based on jump returns strategy.

	Panel A: abnormal overnight return		Panel B: abnormal intraday return		Panel C: abnormal close-to-close return	
decile_jr	alpha	t-stat.	alpha	t-stat.	alpha	t-stat.
1	-0.0062***	-18.23	0.0002	1.20	-0.006***	-16.27
2	-0.0023***	-9.30	-0.0001	-0.34	-0.0024***	-8.36
3	-0.0017***	-6.86	0.0001	0.49	-0.0015***	-5.02
4	-0.0016***	-6.73	0.0009***	3.53	-0.0007**	-2.55
5	-0.0013***	-5.96	0.001***	2.77	-0.0002	-0.82
6	-0.0013***	-5.99	0.0008***	3.63	-0.0005*	-1.66
7	-0.0009***	-4.12	0.0012***	4.17	0.0002	0.90
8	-0.001***	-4.20	-4.20	3.16	0.0001	0.47
9	-0.0001	-0.51	0.0009**	1.82	0.0007**	2.09
10	0.0071***	15.48	-0.0001	-1.28	0.007***	14.76
top-bottom_jr	0.0133***	25.34	-0.0004**	-1.95	0.0129***	20.73

Note: deciles_jr means that stocks are divided into decile groups with jump return (jr); top-bottom_jr represents the stock portfolio with the highest jump return (decile_jr_10) minus the stock portfolio with the lowest cumulative jump return (decile_jr_1); Alpha is the abnormal overnight (or

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intraday or close-to-close) return of the value-weighted stock portfolio grouped by daily jump return after Fama-French-Carhart four-factor risk adjustment; "*, **, ***" respectively represent the coefficient value of 10%, 5 %, 1% confidence level are significant, and the t statistic value in parentheses is the value adjusted by Newey-West.

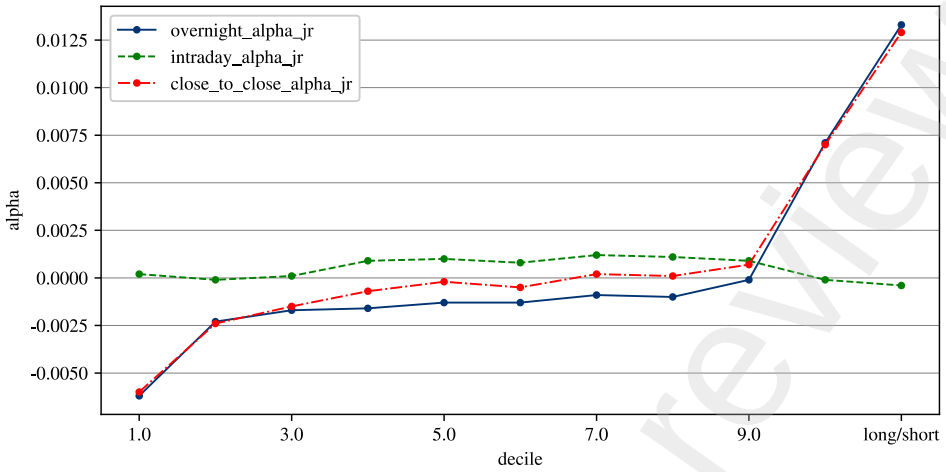


Fig 2. Alphas of portfolios of overnight, intraday return and close-to-close return. Where jr represents jump return.

3.3 Information shock, firm characteristics, and components of return

This section analyzes whether the predictability of information shock on components of return would be affected by firm characteristics. In order to exclude the interference of other conventional pricing factors, we use the components of return adjusted by Fama-French three-factor for analysis:

$$Adj-or_{i,t} = (r_{i,t} - rf_t) - (b_{i,t}MKT_t + c_{i,t}SMB_t + d_{i,t}HML_t) \quad (14)$$

Where, $b_{i,t}$、$c_{i,t}$ and $d_{i,t}$ are estimated by rolling regression of daily data of stock i in the window for the past one year. In the following analysis, $Adj\text{-}or_{i,t}$ is used to replace original overnight return, original intraday return and original close-to-close return.

3.3.1 Information shock and liquidity premium

Extant studies (Amihud, 2002; Lou & Shu, 2017) have found a positive relation between illiquidity and expected return. Also, numerous literatures attribute the short-term price reversal to liquidity or market microstructure effect (Jegadeesh and Titman, 1995; Gutierrez and Kelly, 2008). In addition, Jiang and Kevin (2017) point out that one challenge in testing investor's short-term mis-reactions to large information shocks is to differentiate the information effect from the liquidity effect in short-term stock returns. Thus, in this section, we use the double-sort portfolio analysis to estimate the components of Fama-French-Carhart alphas based on the jump return and the Amihud liquidity factor.

Table 3 shows the adjusted overnight (intraday/close-to-close) alpha of the double-sort stock portfolios based on the jump return and the Amihud illiquidity. Figure 3 plots the corresponding graph.

Specifically, in each past one day, we first divide stocks into quantiles based on their Amihud liquidity measure (*Illiq)*. Then, stocks in each quantile are sorted into quantile based on the lagged jump returns. The results are based on value-weighted portfolio. Panel A and Panel C of Table 3 present that for jump return long-short strategies, the overnight and close-to-close Fama-French-Carhart alphas remain

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statistically significant positive for all different illiquidity subsamples, which indicates that the liquidity does not attribute to explaining the overnight premium involving large information shocks. In other words, we verify that the short-term stock overnight return is truly influenced by information effects rather than liquidity effects.

However, Panel B of Table 3 presents that for jump return long-short strategies, the intraday Fama-French-Carhart alphas is statistically significant negative (-0.67% per day, *t*-statistic of -9.943) for bottom illiquidity subsample while they are not statistically significant in other illiquidity subsamples. These results indicate that the intraday premium of short-term overreaction to large information shock could not be absorbed by the least illiquid stocks. However, the intraday premium is not pervasive of more liquid stocks. Overall, the results indicate that the intraday premium to large information shocks attributes to the liquidity effect to some extent.

Table 3

Abnormal overnight (intraday/close-to-close) return based on jump returns & Amihud sorted strategy.

	Panel A: Abnormal overnight return				
Quintile Sort on jr	Quintile sort on Amihud illiquidity				
	Low	2	3	4	High
Q1	-0.0037***	-0.0033***	-0.0034***	-0.0035***	-0.009***
Q5	-0.0016***	-0.0007**	-0.0005	0.0018***	0.0159***
Q5-Q1	0.0033***	0.0024***	0.0026***	0.0045***	0.0237***
(t-stat)	(5.999)	(5.833)	(6.094)	(9.312)	(22.029)
	Panel B: Abnormal intraday return				
	Quintile sort on Amihud illiquidity				
Sort on jr	Low	2	3	4	High
Q1	0.0006	0.0009**	0.0004	0.0	0.003***
Q5	0.0012**	0.0003	0.0005	-0.0003	-0.0038***
Q5-Q1	0.0006	-0.0008	0.0004	-0.0004	-0.0067***
(t-stat)	(0.701)	(-1.38)	(0.734)	(-0.749)	(-9.943)
	Panel C: Close-to-close return				
	Quintile sort on Amihud illiquidity				
Sort on jr	Low	2	3	4	High
Q1	-0.003***	-0.0023***	-0.0029***	-0.0034***	-0.0059***
Q5	-0.0004	-0.0003	0.0001	0.0016***	0.0122***
Q5-Q1	0.0038***	0.0015**	0.003***	0.0041***	0.017***
(t-stat)	(3.89)	(2.26)	(4.97)	(6.28)	(16.30)

Note: this table reports the abnormal overnight (intraday/close-to-close) return of the value-weighted stock portfolio grouped by jump return and liquidity after Fama-French-Carhart four-factor risk adjustment; jr represents jump return, which is the measure of information shock. Amihud represents a measurement of liquidity; "*, **, ***"respectively represent the coefficient value of 10%, 5 %, 1% confidence level are significant, and the t statistic value in parentheses is the value adjusted by Newey-West.

3.3.2 Information shock and market size effect

According to Fama & French (1996; 2015; 2016), size factor is a systemic risk factor that needs to be priced. This section constructs a double-sort combination based on jump return and stock capitalization to analyze the impact of effect size on the relation between jump return and components of returns.

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Panel A of Table 4 is the 5*5 two-factor grouping of stock market *size* with jump return, see Figure 4 for the intuitive diagram. The results of Panel A and Panel C show that, for all five size subsamples, going long the top quantile of jump return and short the corresponding bottom quantile generate a significantly positive Fama-French-Carhart overnight/close-to-close alpha.

Table 4

Abnormal overnight (intraday/close-to-close) return based on jump returns & Size sorted strategy.

	Panel A: Abnormal overnight return				
Quintile Sort on jr	Quintile sort on size				
	Low	2	3	4	High
Q1	-0.0054***	-0.0054***	-0.0054***	-0.0042***	-0.003***
Q5	0.004***	0.0052***	0.004***	0.003***	0.0017***
Q5-Q1	0.0091***	0.0097***	0.009***	0.0069***	0.0046***
(t-stat)	(14.786)	(14.961)	(12.631)	(11.273)	(9.181)
	Panel B: Abnormal intraday return				
	Quintile sort on size				
Sort on jr	Low	2	3	4	High
Q1	0.0019***	0.0018***	0.0007*	0.0012***	0.0
Q5	0.0004	-0.001**	-0.0007	-0.0002	0.0
Q5-Q1	-0.0015***	-0.0027***	-0.0016***	-0.0014**	0.0001
(t-stat)	(-3.014)	(-4.467)	(-2.883)	(-2.507)	(0.148)
	Panel C: Close to close return				
	Quintile sort on size				
Sort on jr	Low	2	3	4	High
Q1	-0.0034***	-0.0035***	-0.0046***	-0.003***	-0.0028***
Q5	0.0045***	0.0042***	0.0034***	0.0028***	0.0018***
Q5-Q1	0.0075***	0.007***	0.0073***	0.0055***	0.0047***
(t-stat)	(10.94)	(9.50)	(10.02)	(7.72)	(7.24)

Note: this table reports the abnormal overnight (intraday/close to close) return of the value-weighted stock portfolio grouped by jump return and Market Capitalization after Fama-French-Carhart four-factor risk adjustment; jr represents jump return, which is the measure of information shock. "*, **, ***"respectively represent the coefficient value of 10%, 5 %, 1% confidence level are significant, and the t statistic value in parentheses is the value adjusted by Newey-West.

Panel B of Table 4 shows that the long-short strategy based on jump return generates statistically significant negative Fama-French-Carhart intraday alphas for the bottom to the fourth size subsamples. Although Fama-French-Carhart intraday alpha is positive for the top size sample, it is not statistically significant. Those results further confirm that the market size effect plays no role in explaining the intraday premium.

To conclude, the size effect fails to absorb both the overnight and intraday abnormal returns brought about by information shock, which is measured by jump returns.

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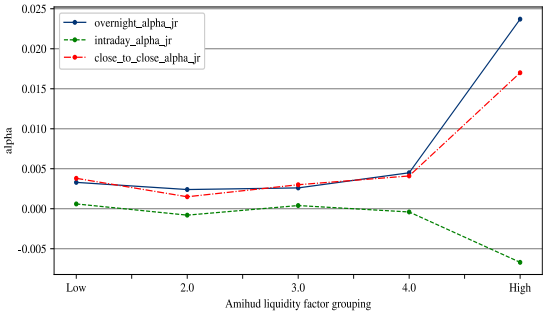


Fig3. Overnight, intraday and close-to-close alphas based on jump returns & Amihud strategy. Where jr represents jump return.

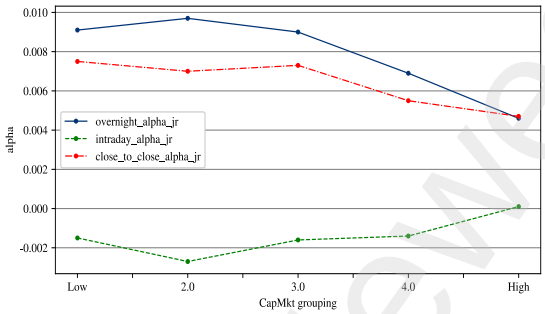


Fig4. Overnight, intraday and close-to-close alphas based on jump returns & size strategy. Where jr and CapMkt represent jump return and stock capitalization respectively.

3.4. Fama-MacBeth regression (overnight/intraday/close-to-close return)

we next turn to Fama-MacBeth regression to explore the cross-section relation between large information shock and subsequent components of returns (that is, overnight return, intraday return and close-to-close return) with other control variables both at the individual stock level and portfolio level.

3. 4.1 Individual stock level

We employ the Fama-MacBeth regressions of the components of return (i.e., overnight return, intraday return and close-to-close return) on (positive/negative) jump return, market sentiment, and other firm characteristics.

In particular, volume is the proxy for market sentiment as high trading volume means high investor sentiment while low trading volume means low investor sentiment to some extent (Chen et al., 2014). In Table 5, where three target returns are overnight return, intraday return and close-to-close return, see Model 2 (M2), Model 4 (M4) and Model 6(M6), respectively. Model 1, Model 3 and Model 5 are the corresponding models for Model 2, Model 4 and Model 6 without control variables.

From M1 and M3, the coefficients of Lag_jr are significantly positive, confirming a positive relation between jump return and the subsequent stock overnight/close-to-close returns, demonstrating that the positive traditional (close-to-close) return-jump risk tradeoff mainly comes from the non-trading period, that is, it is the overnight component rather than the intraday component of close-to-close return that contributes to the return-higher-order moment risk tradeoff. From M2 and M6 we can conclude that after controlling for other firm characteristics and market investors' sentiment, the positive return-jump tradeoff at night still stands robust.

Table 5

Fama-MacBeth regression of individual stock level.

	M1(Overnight)	M2	M3(Intraday)	M4	M5(Close-to-close)	M6
Intercept	-0.0362***	-0.1538***	0.0218***	-0.0892***	-0.0121***	-0.2161***
	(-10.26)	(-9.14)	(6.76)	(-4.59)	(-7.81)	(-11.97)
Lag_jr	0.0164***	0.0158***	0.0155**	0.0105	0.0318***	0.0279***
	(3.48)	(4.16)	(2.23)	(1.53)	(4.03)	(3.94)
Lag_pos_jr	-0.0171***	-0.0072	0.0256***	0.0117	0.0084	0.0071

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	(2.74)	(-1.48)	(3.55)	(1.61)	(0.94)	(0.78)
Lag_neg_jr	0.0334***	0.0231***	-0.0101*	-0.0011	0.0234***	0.0208***
	(5.90)	(4.35)	（-1.67）	(-0.18)	(3.21)	(2.77)
Size		0.0131***		-0.0097***		0.0026**
		(16.39)		(-8.62)		(1.96)
BM		0.0001***		-0.0001***		-0.0000
		(3.53)		(-5.68)		(-1.33)
Liquid_m		0.0382***		-0.0014		0.0469***
		(3.36)		(-0.13)		(4.15)
Volume		-0.0096***		0.0176***		0.0078***
		(-9.95)		(11.28)		(4.45)

Note: Lag_jr, Lag_pos_jr and Lag_neg_jr represent jump return, positive jump return and negative jump return with one-period lag respectively. Size is the market value of the stock, Liquid_m is the Amihud liquidity of the stock, BM is the book-to-market ratio of the stock, and volume is the trading volume of the stock. "*, **, ***" respectively represent the coefficient value of 10%, 5 %, 1% confidence level are significant and the value in parentheses is t-statistic value.

3.4.2 Portfolio level

In this part, we continue to carry on Fama-MacBeth regression at portfolio level. Stocks are divided into 10 portfolios according to one-period lagged jump return, positive jump return and negative jump return every month. The value-weighted average overnight/intraday/close-to-close return and the simple one-period lagged average jump return, positive jump return and negative jump return of the constituent stocks in the portfolios are calculated every month.

From Panel A and Panel C of Table 6, the coefficients of port_jr/port_neg_jr are significantly positive and the coefficients of port_pos_jr are significantly negative after controlling Fama-French three factors, confirming a positive relation between jump return/negative jump return and the subsequent stock overnight/close-to-close returns and a negative relation between positive jump return and the subsequent stock overnight/close-to-close returns. However, the results in Panel B show a positive relation between positive jump return and the subsequent stock intraday returns and a negative relation between negative jump return and the subsequent stock intraday returns. These findings demonstrate that positive traditional (close-to-close) return-jump risk tradeoff mainly comes from the non-trading period(overnight) rather than trading period(intraday).

Table 6

Fama-MacBeth regression of portfolio level.

Panel A: Overnight return			
	M1	M2	M3
Intercept	-0.0357***	-0.0335***	-0.0333***
	(-10.31)	(-9.61)	(-9.58)
port_jr	0.0291***		
	(4.87)		
port_pos_jr		-0.0705***	
		(-5.58)	
port_neg_jr			0.0794***
			(9.53)
Mkt_rf	0.0002	0.0002	0.0002
	(0.28)	(0.24)	(0.27)
SMB	-0.0004***	-0.0004**	-0.0004**
	(-2.66)	(-2.42)	(-2.54)
HML	-0.0001	-0.0001	-0.0001

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	(-0.57)	(-0.58)	(-0.46)
Panel B: Intraday return			
	M1	M2	M3
Intercept	0.0218***	0.0212***	0.0205***
	(6.88)	(6.36)	(6.08)
port_jr	0.0172		
	(1.36)		
port_pos_jr		0.0504***	
		(3.35)	
port_neg_jr			-0.0220*
			(-1.70)
Mkt_rf	-0.0000	0.0001	-0.0001
	(-0.01)	(0.10)	(-0.12)
SMB	0.0001	0.0002	0.0001
	(0.94)	(1.02)	(0.91)
HML	0.0002*	0.0002*	0.0002*
	(1.76)	(1.72)	(1.91)
Panel C: Close-to-close return			
	M1	M2	M3
Intercept	-0.0117***	-0.0101***	-0.0106***
	(-6.94)	(-6.40)	(-6.64)
port_jr	0.0463***		
	(4.48)		
port_pos_jr		-0.0201*	
		(-1.85)	
port_neg_jr			0.0575***
			(3.57)
Mkt_rf	0.0002	0.0002	0.0001
	(0.63)	(0.76)	(0.41)
SMB	-0.0003***	-0.0002**	-0.0002***
	(-3.21)	(-2.50)	(-3.14)
HML	0.0001**	0.0001**	0.0002***
	(2.18)	(2.13)	(2.97)

Note: port_jr, port_pos_jr and port_neg_jr represent jump return, positive jump return and negative jump return of portfolio with one-period lag respectively. Mkt_rf, SMB and HML are Fama-French three factors. "*, **, ***" respectively represent the coefficient value of 10%, 5 %, 1% confidence level are significant and the value in parentheses is t-statistic value.

4. Further analysis

Existing studies have confirmed that positive and negative jump risk measurements have an asymmetric effect on stock returns (Rabemananjara and Zakoian, 1993; Guo et al., 2013). Thus, we further calculate the daily positive and negative cumulative jump returns based on positive and negative jump risk measurements to analyze the positive and negative jump returns-based trading strategies.

4.1 Asymmetric information shock and components of return

Panel A, Panel B and Panel C of Table 7 decompose the abnormal return on a positive jump return strategy. Going long the high positive jump return decile and short the low positive jump return decile, we find that the positive jump return premium occurs overnight (1.45% per day, *t*-statistic of 23.11) as there are statistically significant negative intraday Fama-French-Carhart alpha (-0.27% per day, *t*-statistic of -3.99) and statistically significant positive close-to-close Fama-French-Carhart alpha (1.19% per day, t-statistic of 14.94). Similarly, Panel D, Panel E and Panel F of Table 7 present that going long (short) the highest (lowest) negative jump return decile can generate a Fama-French-Carhart adjusted overnight premium of 0.99% with an associated *t*-statistic of 27.47 and a Fama-French-Carhart adjusted close-to-close

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premium of 0.88% with an associated t-statistic of 13.41. Although the intraday Fama-French-Carhart adjusted alpha is slightly negative, it is not statistically significant (-0.01% per day, *t*-statistic of -0.13). In all, these results provide evidence of an apparent asymmetric predictability of positive and negative information shock at night, with the adjusted overnight alpha obtained by trading strategy based on positive jump return being almost twice as much as that based on negative jump return, where the values are 1.45% versus 0.88% per day.

Table 7

Abnormal overnight/intraday/close-to-close return based on positive (negative) jump returns strategy

	Panel A: abnormal overnight return		Panel B: abnormal intraday return		Panel C: abnormal close-to-close return	
decile_pos_jr	alpha	t-stat.	alpha	t-stat.	alpha	t-stat.
1	-0.0013***	-5.46	0.0007*	1.70	-0.0006	-1.43
10	0.0133***	20.97	-0.0019***	-3.32	0.0113***	16.16
top-bottom_pos_jr	0.0145***	23.11	-0.0027***	-3.99	0.0119***	14.94
	Panel D: abnormal overnight return		Panel E: abnormal intraday return		Panel F: abnormal close-to-close return	
decile_neg_jr	alpha	t-stat.	alpha	t-stat.	alpha	t-stat.
1	-0.0101***	-21.82	0.0009*	1.93	-0.0092***	-17.07
10	-0.0012***	-5.11	0.0008**	2.24	-0.0004	-1.05
top-bottom_neg_jr	0.0088***	19.66	-0.0001	-0.13	0.0088***	13.41

Note: deciles_pos_jr means that stocks are divided into decile groups with positive jump return (pos_jr) while deciles_neg_jr means that stocks are divided into decile groups with negative jump return (neg_jr); top-bottom_pos_jr represents the stock portfolio with the highest positive jump return (decile_pos_jr_10) minus the stock portfolio with the lowest cumulative positive jump return (decile_pos_jr_1) while top-bottom_neg_jr represents the stock portfolio with the highest negative jump return (decile_neg_jr_10) minus the stock portfolio with the lowest cumulative negative jump return (decile_neg_jr_1); Alpha is the abnormal overnight (or intraday or close-to-close) return of the value-weighted stock portfolio grouped by daily jump return after Fama-French-Carhart four-factor risk adjustment; "*, **, ***" respectively represent the coefficient value of 10%, 5 %, 1% confidence level are significant, and the t statistic value in parentheses is the value adjusted by Newey-West.

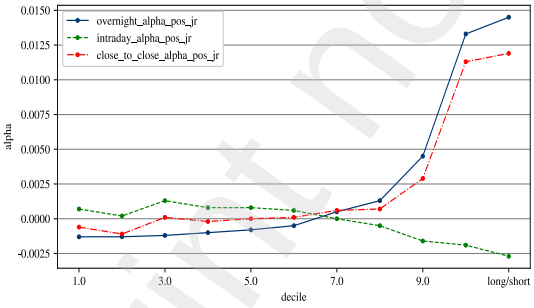


Fig5. Alphas of portfolios of overnight return, intraday return and close-to-close return. Where pos_jr represents positive jump return.

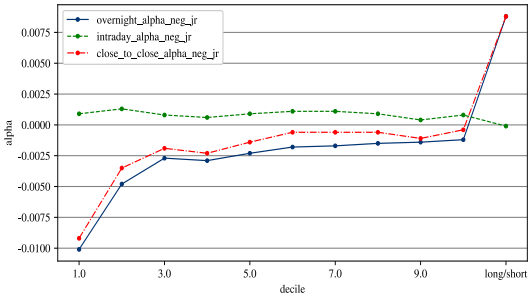


Fig6. Alphas of portfolios of overnight return, intraday return and close-to-close return. Where neg_jr represents negative jump return.

4.2 Asymmetric Information shock, firm characteristics and components of return

Panel A of Table 8 presents that the long-short strategy based on positive jump return generates twice more Fama-French-Carhart overnight alpha than that based on negative jump return for the most illiquidity subsamples, with the statistics of 1.24% (*t*-statistic of 8.305) versus 0.56% (*t*-statistic of 6.066) per day. In addition, Panel B of Table 8 presents that the long-short strategy based on positive jump return

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can generate much more Fama-French-Carhart negative intraday alpha for the most illiquidity subsample with the value of -0.44% (*t*-statistic of -3.758). Panel C of Table 8 presents that the long-short strategy based on positive jump return generates almost twice more Fama-French-Carhart close-to-close alpha than that based on negative jump return for the most illiquidity subsamples, with the statistics of 0.8% (t-statistic of 5.45) versus 0.47% (t-statistic of 3.76) per day.

Panel A of Table 9 show that the long-short strategy based on positive jump return generates greater Fama-French-Carhart overnight alpha than that based on negative jump return for all size subsamples. Panel B of Table 9 show that the long-short strategy based on positive jump return or negative jump return can rarely get Fama-French-Carhart intraday alpha through size subsamples. Panel C of Table 9 show that the long-short strategy based on positive jump return generates greater Fama-French-Carhart close-to-close alpha than that based on negative jump return for the bottom to the fourth size subsamples.

Overall, we further verify that the positive overnight and close to close premium is pervasive of the least liquid stocks and small firms and particularly microcaps. Moreover, it is positive jump return rather than negative jump return that mainly contributes to the positive (negative) overnight (intraday) return due to investors' preference to buying winners in Chinese stock market.

Table 8

Positive (negative) jump return, Amihud illiquidity, and abnormal overnight (intraday/close-to-close) return.

Panel A: Abnormal overnight return					
Quintile Sort on pos_jr	Quintile sort on Amihud illiquidity				
	Low	2	3	4	High
Q1	-0.0015***	-0.0011***	-0.0009***	-0.0004	-0.0005
Q5	0.0006	0.001	0.0017***	0.0039***	0.0186***
Q5-Q1	0.0015*	0.0009	0.0014**	0.0028***	0.0124***
(t-stat)	(1.772)	(1.416)	(2.113)	(4.122)	(8.305)
	Quintile sort on Amihud illiquidity				
Sort on neg_jr	Low	2	3	4	High
Q1	-0.0074***	-0.0057***	-0.0052***	-0.0053***	-0.0103***
Q5	-0.0014***	-0.0012***	-0.0012***	-0.0018***	-0.0031***
Q5-Q1	0.0058***	0.0036***	0.0033***	0.0026***	0.0056***
(t-stat)	(9.105)	(7.877)	(8.743)	(5.335)	(6.066)
Panel B: Abnormal intraday return					
Quintile	Quintile sort on Amihud illiquidity				
Sort on pos_jr	Low	2	3	4	High
Q1	0.0006*	0.0006	0.0004	0.0006	0.0025***
Q5	0.0009	-0.0006	-0.0005	-0.0007	-0.0039***
Q5-Q1	0.0011	-0.0014	0.0002	0.0	-0.0044***
(t-stat)	(1.026)	(-1.592)	(0.260)	(0.010)	(-3.758)
	Quintile sort on Amihud illiquidity				
Sort on neg_jr	Low	2	3	4	High
Q1	0.0017**	0.0017***	0.0013**	0.0006	0.0022***
Q5	0.0003	0.0008*	0.0013***	0.0021***	0.0021***
Q5-Q1	-0.0011	-0.0001	-0.0006	0.0022***	-0.0009
(t-stat)	(-1.220)	(-0.121)	(-0.793)	(2.873)	(-0.922)

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Panel C: Close-to-close return					
Quintile	Quintile sort on Amihud illiquidity				
Sort on pos_jr	Low	2	3	4	High
Q1	-0.0009***	-0.0004	-0.0004	0.0003	0.002**
Q5	0.0015	0.0005	0.0013*	0.0032***	0.0148***
Q5-Q1	0.0026**	-0.0005	0.0015*	0.0028***	0.008***
(t-stat)	(2.09)	(-0.50)	(1.67)	(2.81)	(5.45)
	Quintile sort on Amihud illiquidity				
Sort on neg_jr	Low	2	3	4	High
Q1	-0.0056***	-0.0039***	-0.0038***	-0.0047***	-0.008***
Q5	-0.001***	-0.0003	0.0002	0.0005	-0.0009
Q5-Q1	0.0047***	0.0035***	0.0028***	0.0047***	0.0047***
(t-stat)	(4.43)	(4.29)	(3.72)	(5.93)	(3.76)

Note: this table reports the abnormal overnight (intraday/close-to-close) return of the value-weighted stock portfolio grouped by positive/negative jump return and liquidity after Fama-French-Carhart four-factor risk adjustment; pos_jr and neg_jr represent positive jump return and negative jump return respectively, which are measures of information shock. Amihud represents a measurement of liquidity; "*, **, ***"respectively represent the coefficient value of 10%, 5 %, 1% confidence level are significant, and the t statistic value in parentheses is the value adjusted by Newey-West.

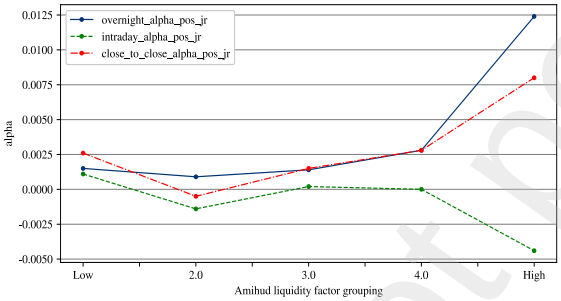


Fig7. Overnight, intraday and close-to-close alphas based on positive jump returns & Amihud strategy. Where pos_jr represents positive jump return.

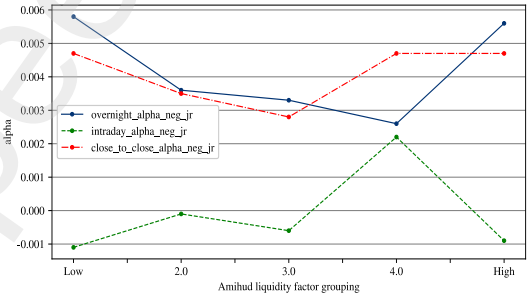


Fig8. Overnight, intraday and close-to-close alphas based on negative jump returns & Amihud strategy. Where neg_jr represents negative jump return.

Table 9

Positive (negative) jump return, size, and abnormal overnight (intraday/close-to-close) return.

Panel A: Abnormal overnight return					
Quintile Sort on pos_jr	Quintile sort on size				
	Low	2	3	4	High
Q1	-0.0017***	-0.0015***	-0.0015***	-0.001***	-0.0011***
Q5	0.01***	0.0094***	0.0093***	0.0069***	0.0057***
Q5-Q1	0.0102***	0.0089***	0.0099***	0.0064***	0.005***
(t-stat)	(11.471)	(9.972)	(10.528)	(7.638)	(7.016)
	Quintile sort on size				
Sort on neg_jr	Low	2	3	4	High
Q1	-0.0079***	-0.0084***	-0.0076***	-0.0062***	-0.0057***
Q5	-0.0024***	-0.0021***	-0.002***	-0.0013***	-0.0011***
Q5-Q1	0.005***	0.0046***	0.0051***	0.0042***	0.0044***

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(t-stat)	(10.449)	(9.024)	(8.520)	(8.210)	(7.882)
Panel B: Abnormal intraday return					
Quintile	Quintile sort on size				
Sort on pos_jr	Low	2	3	4	High
Q1	0.002***	0.0001	0.0006	0.0003	-0.0002
Q5	-0.0017***	-0.0017**	-0.0026***	-0.0017**	-0.0013*
Q5-Q1	-0.0046***	-0.0013	-0.0027***	0.0001	-0.0006
(t-stat)	(-5.578)	(-1.441)	(-3.213)	(0.159)	(-0.738)
	Quintile sort on size				
Sort on neg_jr	Low	2	3	4	High
Q1	0.0018***	0.0022***	0.0019***	0.002***	0.0005
Q5	0.0026***	0.0015***	0.0008*	0.0003	0.0004
Q5-Q1	0.0005	-0.0003	-0.0005	-0.0016***	0.0001
(t-stat)	(0.737)	(-0.448)	(-0.770)	(-2.357)	(0.230)
Panel C: Close-to-close return					
Quintile	Quintile sort on size				
Sort on pos_jr	Low	2	3	4	High
Q1	0.0003	-0.0013***	-0.0008*	-0.0006	-0.0012***
Q5	0.0084***	0.0078***	0.0067***	0.0053***	0.0044***
Q5-Q1	0.0056***	0.0076***	0.0072***	0.0065***	0.0043***
(t-stat)	(6.10)	(7.54)	(6.92)	(6.06)	(4.54)
	Quintile sort on size				
Sort on neg_jr	Low	2	3	4	High
Q1	-0.006***	-0.006***	-0.0056***	-0.0041***	-0.0051***
Q5	0.0002	-0.0005	-0.0011***	-0.0009**	-0.0006
Q5-Q1	0.0054***	0.0043***	0.0045***	0.0026***	0.0045***
(t-stat)	(7.77)	(5.74)	(5.58)	(3.52)	(5.34)

Note: this table reports the abnormal overnight (intraday/close-to-close) return of the value-weighted stock portfolio grouped by positive/negative jump return and Market Capitalization after Fama-French-Carhart four-factor risk adjustment; pos_jr and neg_jr represent positive jump return and negative jump return respectively, which are measures of information shock. "*, **, ***"respectively represent the coefficient value of 10%, 5 %, 1% confidence level are significant, and the t statistic value in parentheses is the value adjusted by Newey-West.

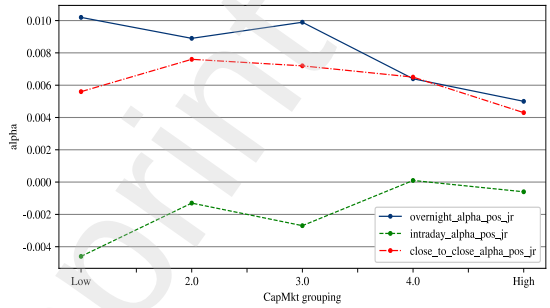


Fig9. Overnight, intraday and close-to-close alphas based on positive jump returns & size strategy. Where pos_jr and CapMkt represent positive jump return and stock capitalization respectively.

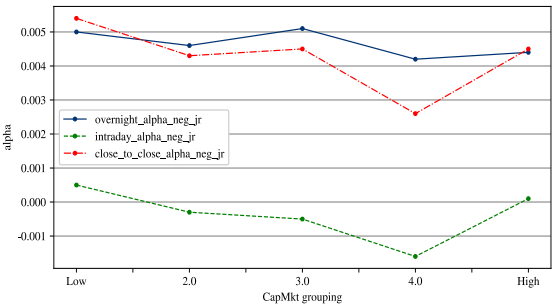


Fig10. Overnight, intraday and close-to-close alphas based on negative jump returns & size strategy. Where neg_jr and CapMkt represent negative jump return and stock capitalization respectively.

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6. Conclusion

This paper uses jumps in stock prices as the proxy for major information shock to examine the relation between jump return and three kind of returns (close-to-close return and its two components, namely overnight return and intraday return) in the Chinese stock market. Taking daily frequency and high-frequency 5-minute individual stock trading data as samples, we firstly construct trading strategies based on jump return to investigate whether they can obtain statistically significant close-to-close return, overnight returns and intraday returns. Moreover, we turn to Fama and MacBeth (1973) regressions to describe the cross-section of the three kinds of expected returns.

Specifically, the empirical results demonstrate a positive relation between jump return and the subsequent stock overnight/close-to-close returns, confirming that the positive risk-return tradeoff originates from the overnight period. This relation also holds after controlling for stock characteristics and investor sentiment.

Finally, we further analyze whether the positive jump return and negative jump return have an asymmetric effect on overnight and intraday components of stock returns. We provide evidence of an apparent asymmetric positive (negative) overnight (intraday) return to positive and negative information shock at night (during the day). In essential, it is positive jump return rather than negative jump return that contributes to the positive trend of jump return and we could attribute this to investors' favor in "buying the winner", which is a unique phenomenon in Chinese stock market.

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